

■ Deloitte OA Preparation Book (Analyst – T&T; Role)

Complete Notes + Practice Questions + Solutions

1. Quantitative Aptitude (Formulas + Shortcuts)

- Percentages: $\% \text{ change} = \frac{(\text{New} - \text{Old})}{\text{Old}} \times 100$
- Averages: $\text{Avg} = \frac{\text{Sum}}{n}$; $\text{New Avg} = \frac{\text{Old Avg} \times n + x}{n+1}$
- Speed = Distance/Time; Train Time = (Train+Object)/Speed
- Probability = Favourable/Total
- SI = $\frac{P \times R \times T}{100}$; CI = $P(1 + \frac{R}{100})^T - P$

2. Logical Reasoning (Key Points)

- Blood Relations: Draw family tree; track generation levels
- Seating Arrangement: Fix one person; arrange others relative
- Syllogisms: Use Venn Diagrams
- Data Sufficiency: Don't solve fully, just check sufficiency

3. Verbal Ability (Common Errors)

- Tenses: 'did not went' ■ → 'did not go' ■
- Subject-Verb: 'He have' ■ → 'He has' ■
- Para Jumbles: Look for connectors (however, therefore, firstly)
- Reading Comprehension: Skim questions first → scan passage

4. Technical (SQL, Python, Statistics)

- SQL: `SELECT dept, COUNT(*) FROM employees GROUP BY dept;`
- SQL (2nd highest salary): `SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);`
- Python: Reverse string `s[::-1]`
- Python: Dict comprehension `{x:x**2 for x in range(5)}`
- Stats: Mean= $\frac{\text{sum}}{n}$, Median=middle, Mode=most frequent, Variance= avg sq.diff , StdDev= $\sqrt{\text{Variance}}$

5. Full Practice Mock Test (25Q + Solutions)

Q1. A train 240m long crosses a platform 360m long in 30s. Find speed in km/hr.

Ans: $(240+360)/30=20$ m/s= 72 km/hr

Q2. A can complete work in 10 days, B in 15 days. Together?

Ans: $(1/10+1/15)^{-1} = 6$ days

Q3. If 40% of a number is 160, find the number.

Ans: 400

Q4. A:B income ratio=3:5, expenses ratio=2:3. A saves 600, B saves 1000. Find A's income.

Ans: A=2400

Q5. Bag with 4 red, 6 blue. P(red)?

Ans: $4/10=0.4$

Q6. SI on 1200 at 10% for 2 yrs?

Ans: 240

Q7. Average of first 15 natural numbers?

Ans: $(1+15)/2=8$

Q8. Speed downstream=12, upstream=8. Still water?

Ans: $(12+8)/2=10$ km/hr

Q9. All books are papers. Some papers are pens. → Which conclusion follows?

Ans: Only (ii): Some papers are books

Q10. Rahul: 'He is son of my grandfather's only son.' Relation?

Ans: Brother

Q11. Ravi 15th left, 10th right. Total?

Ans: 24

Q12. Seating: C A B D E → Who sits middle?

Ans: B

Q13. Code 'CAB' if A=1, B=2, C=3.

Ans: 312

Q14. Series 2,6,12,20,30,...?

Ans: 42

Q15. Error: 'Neither of the boys have finished'

Ans: Replace 'have' with 'has'

Q16. Synonym of 'Mitigate'

Ans: Lessen

Q17. Arrange: (B-A-C-D)

Ans: B→A→C→D

Q18. Cause of global warming?

Ans: Human activities

Q19. Antonym of 'Optimistic'

Ans: Pessimistic

Q20. Fill: He is very good ____ Mathematics.

Ans: at

Q21. SQL: employees earning > avg salary

Ans: `SELECT * FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);`

Q22. SQL: 2nd highest salary

Ans: `SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);`

Q23. Python: reverse 'hello'

Ans: `'hello'[::-1]`

Q24. Python two-sum indices for target=9 in [2,7,11,15]

Ans: (0,1)

Q25. Mean of [2,4,4,4,5,5,7]

Ans: 4.43